Relativistic Stark resonances in Debye plasmas: screening undoes the relativistic suppression of ionization widths

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The three ingredients of a hydrogen-like ion in a dense plasma under a strong static field — relativity, Debye screening, and field ionization — have each been treated in pairs; nonperturbative resonance parameters of the combined problem have, to our knowledge, never been reported. We compute them: positions and widths of the Dirac equation with a Yukawa-screened Coulomb potential and a uniform electric field, by complex scaling of a coupled κ-channel system whose radial channels carry the exact-structure Laguerre bases of our companion program. The engine is validated on a ladder — per-channel Sommerfeld spectra to 10^{-11}, the relativistic Stark benchmarks of Maltsev *et al.* to seven digits, and the nonrelativistic screened-Stark limit against an independent engine to 4×10^{-6} — and its spurious $\kappa > 0$ intruder, exactly degenerate with the $1s$ level at zero screening, is identified and excluded per channel using the pollution theory of our companion theorem paper. Two results follow. First, in the scaled units $(\lambda/Z,\, F/Z^3)$ in which nonrelativistic physics collapses to a single curve, the relativistic width suppression discovered at zero screening weakens monotonically as screening grows: the suppression factor rises by 0.2% at $Z = 10$ but by 62% at $Z = 82$, so the plasma catalyzes field ionization of a heavy ion 1.6 times more strongly than nonrelativistic scaling predicts — screening lifts the level toward threshold and deflates the exponential sensitivity of the tunneling width to the relativistic binding correction, which itself survives nearly intact. Second, resolving the $n = 2$, $m = 1/2$ manifold at $Z = 40$ into its three field-dressed components, the outer pair is born with equal widths where the matched nonrelativistic doublet is thirty-fold asymmetric, and its width ratio is non-monotonic in the screening parameter — falling from 1.02 to 0.22 and recovering to 0.33 while the nonrelativistic ratio rises monotonically — as the uphill component crosses $E_r = 0$ and persists as a positive-energy relativistic resonance.

I. INTRODUCTION

Each pairwise combination of relativity, plasma screening, and static fields in a hydrogen-like ion is by now classical. The relativistic Stark problem — the Dirac equation in a uniform field — was solved nonperturbatively by complex rotation for low Z [1] and recently mapped across $Z = 1$–82 by Maltsev *et al.* [2], whose central finding is a *relativistic suppression* of ionization widths reaching an order of magnitude at high Z. The screened Stark problem at the nonrelativistic level has a two-decade lineage [3–5], extended by our companion map of the (F, λ) plane [6]. The screened Dirac problem without a field is likewise developed, from bound levels [7] to resonances [8, 9]. The triple combination — Dirac, screening, and field at once — appears never to have been computed at the level of nonperturbative resonance positions and widths (the three ingredients do meet in the adjacent line-broadening literature [10, 11], where the computed quantity is collisional electron-impact broadening rather than static-field ionization), although it is the natural setting for highly charged ions in dense plasmas, and although the obvious question writes itself: does Maltsev's relativistic suppression survive when the plasma screens the Coulomb attraction?

This paper computes the triple and answers the question: the suppression does not fully survive. Screening lifts the level toward the continuum threshold; the relativistic deepening of the binding survives almost intact, but the exponential sensitivity of the tunneling width to that offset deflates, and the width recovers toward its nonrelativistic value — weakly at low Z, dramatically in the heavy-ion regime. In a plasma, a highly charged ion is therefore doubly fragile: the familiar screening catalysis of field ionization [3, 6] is compounded by the removal of the relativistic width protection.

The tool is a coupled-channel complex-scaling method assembled from two validated components of our program: the exact-structure radial Dirac bases (Coulomb–Dirac cusp exponent r^γ, certified assembly) [9, 12] and the coupled-channel Stark machinery of the nonrelativistic companion [6]. The one genuinely new element — the relativistic dipole coupling between κ-channels — is certified by two independent derivations. A known hazard of finite-basis Dirac calculations turns out to matter operationally: the spurious $\kappa > 0$ state that is *exactly* degenerate with the physical $1s$ level at zero screening [13, 14] detaches downward under screening and, if untreated, silently replaces the physical reference. Our companion pollution theory [15] identifies it; per-channel referencing excludes it by construction.

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II. METHOD

A. Coupled κ-channels

For a point nucleus of charge Z with Yukawa screening and a field $F\hat{z}$ (atomic units, $c = 137.035999$),

$$H = c\,\boldsymbol{\alpha}\cdot\mathbf{p} + \beta c^2 - \frac{Ze^{-\lambda r}}{r} + Fz, \tag{1}$$

where λ is the inverse Debye length of the plasma, the magnetic quantum number m is conserved; we work in the lowest sector $m = 1/2$. Each relativistic angular channel, labeled by the Dirac quantum number $\kappa = \mp(j+\frac{1}{2})$ for $j = \ell \pm \frac{1}{2}$ [the eigenvalue of $-(1+\boldsymbol{\sigma}\cdot\mathbf{L})$], contributes large and small radial components; the field couples channels through the angular matrix elements of $\cos\theta$ between spin-angular functions, $A^L_{\kappa'\kappa} = \langle\Omega_{\kappa' m}|\cos\theta|\Omega_{\kappa m}\rangle$ for the large components and $A^S_{\kappa'\kappa} = A^L_{-\kappa',-\kappa}$ for the small ones, evaluated from the Wigner–Eckart reduced matrix element [16]. These coefficients were certified by an independent Clebsch–Gordan/Gaunt decomposition, the two derivations agreeing to machine zero on all pairs; their s-limit reproduces the nonrelativistic Gaunt factor exactly.

Radially, every channel carries the orthonormal exact-structure Laguerre basis of Ref. [12] (r^{γ_κ} prefactor, $\gamma_\kappa = \sqrt{\kappa^2 - (Z/c)^2}$) on a *shared* quadrature grid, so cross-channel dipole integrals $\langle f_{\kappa'}|r|f_\kappa\rangle$ are exact contractions; production settings use twelve channels ($\kappa = \pm1, \pm2, \ldots, \pm6$), $n_b = 52$ functions per channel and component ($2 \times 52 \times 12 = 1248$ degrees of freedom), scale $\beta = 0.55Z$, and overlap/integration-by-parts certifications at the 10^{-12}–10^{-8} level across $Z = 1$–82. The per-channel diagonal blocks are the mass-shifted complex-symmetric Dirac pencils of Ref. [9] under the complex rotation $r \to re^{i\theta_c}$ (screening evaluated at complex argument), which exposes each resonance as an isolated complex eigenvalue $E = E_r - i\Gamma/2$, with E_r the position and Γ the full ionization width; the field blocks are $Fe^{i\theta_c}A^{L,S}_{\kappa'\kappa}\langle f_{\kappa'}|r|f_\kappa\rangle$. For the *nonrelativistic* screened-Stark operator this rotation rests on dilation-analyticity theory [17–19]; no analogous theorem is known for the Dirac operator with a linear potential, and, as in the relativistic Stark literature [1, 2], complex rotation is applied here as a computational tool [20, 21] whose operational justification is the two-angle acceptance test and the validation ladder below.

B. The impostor and its exclusion

Finite-basis Dirac discretizations of the $\kappa > 0$ channels carry a spurious state exactly degenerate with the physical $1s$ ($-\kappa$) level [13, 14]. Under screening this degeneracy is lifted the wrong way: the impostor detaches *downward*, and a full-matrix lowest-eigenvalue selector then anchors all continuations to an unphysical, basis-dependent state — we observed exactly this failure, flagged because the selected level fell *below* the variational bound implied by the hydrogenic scaling $E(Z,\lambda) = Z^2E(1,\lambda/Z)$. The standard cure is a dual-kinetic-balance basis [22]; here a lighter fix suffices: all field-free references are taken from the corresponding κ-channel block alone (the physical $1s$ from $\kappa = -1$, where no impostor exists), which excludes the intruder by construction; the mechanism and its per-fiber theory are the subject of the companion theorem paper [15].

C. Identification, scaling, uncertainties

Resonances are identified as in Refs. [6, 9]: parametric continuation (the search window predicted from the channel-block field-free level plus the previous complex field shift), two-angle acceptance with the pairing rule of Ref. [6] and the modulus difference recorded as the spread. The angle pairs are chosen per Z as the smallest rotations that expose the resonance while remaining θ-stable — the relativistically narrowed high-Z resonances require smaller angles: $\theta_c = 0.14/0.22$ at $Z = 10$; $0.10/0.16$ at $Z = 20$ and 40 (including the $n = 2$ manifold); $0.08/0.13$ at $Z = 54$ and 82; the $Z = 1$ validation runs used $0.18/0.26$. We present results in the scaled variables $u = \lambda/Z$ and $f = F/Z^3$, in which nonrelativistic physics is Z-independent ($E = Z^2\tilde{E}$, $\Gamma = Z^2\tilde{\Gamma}$): every deviation from the collapse is pure relativity. The nonrelativistic baseline $\tilde{\Gamma}_{\rm nr}(u,f)$ is computed live with the engine of Ref. [6]. Quoted uncertainties are two-angle spreads — estimates calibrated on the validation ladder, not bounds. The dominant systematic is channel truncation, calibrated at 1.4×10^{-3} relative in widths at the production twelve-channel setting [step (ii) below], so tabulated suppression factors are significant to about three digits.

D. Validation ladder

(i) Field-free, per channel: exact Dirac–Coulomb (Sommerfeld) energies reproduced at 10^{-11} ($Z = 1$–40) to 10^{-8} ($Z = 92$, $s_{1/2}$). (ii) Stark, no screening: the benchmarks of Ref. [2] at $F/Z^3 = 0.05$ reproduced at $Z = 1, 10, 18$; at converged settings (20 channels, $n_b = 64$) the $Z = 1$ shift agrees to 5×10^{-8} and the width to 2×10^{-6} relative. (iii) Screening, low Z: at $Z = 1$, $F = 0.10$, $\lambda = 0.20$ the resonance agrees with the independently written nonrelativistic engine of Ref. [6] to 4×10^{-6} — the residual being the $1s$ Dirac shift itself. (iv) The $u = 0$ suppression factors below reproduce the ratios implied by Maltsev's tables (e.g. $S = 0.4089$ at $Z = 54$ versus 0.4088).

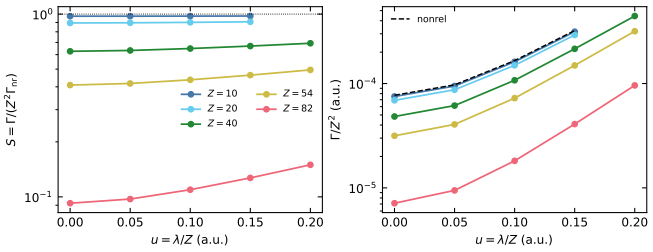


FIG. 1. Left: relativistic suppression factor $S = \Gamma/(Z^2\Gamma_{\rm nr})$ of the $1s$ Stark resonance at $f = F/Z^3 = 0.05$ versus scaled screening $u = \lambda/Z$, for $Z = 10$–82 (log scale). Screening drives every curve toward the nonrelativistic limit $S = 1$ (dotted). Right: scaled widths Γ/Z^2; the dashed curve is the nonrelativistic baseline, on which all Z would collapse in the absence of relativity.

TABLE I. Suppression factor $S(Z, u)$ of the $1s$ width at $f = 0.05$. Two-angle spreads below 5×10^{-4} (absolute; worst corner 4.4×10^{-4} at $Z = 54$, $u = 0.20$). The $u = 0$ column reproduces the field-free-screening ratios implied by Ref. [2].

Z	$u = 0$	0.05	0.10	0.15	0.20
10	0.9737	0.9740	0.9747	0.9760	—
20	0.8943	0.8963	0.9011	0.9076	—
40	0.6258	0.6324	0.6479	0.6685	0.6924
54	0.4089	0.4173	0.4369	0.4635	0.4950
82	0.0923	0.0973	0.1094	0.1270	0.1497

III. RESULTS

A. Screening undoes the relativistic width suppression

Figure 1 and Table I present the central result: the suppression factor $S(Z, u) = \Gamma/(Z^2\tilde{\Gamma}_{\rm nr})$ of the $1s$ Stark resonance at fixed scaled field $f = 0.05$, as screening grows from $u = 0$ to 0.20.

Every row rises: screening always weakens the relativistic suppression, and the effect grows steeply with Z — a 0.2% recovery at $Z = 10$, +10.6% at $Z = 40$, +21% at $Z = 54$, and +62% at $Z = 82$ over the computed range. The mechanism is not an erosion of the relativistic binding itself: the scaled relativistic deepening $\Delta(u) = |E_0|/Z^2 - |E_0^{\rm nr}|$ of the reference level is nearly independent of screening ($0.0552 \to 0.0519$ at $Z = 82$; $0.0111 \to 0.0104$ at $Z = 40$), because the Debye potential is a long-range modification ($e^{-\lambda r}/r \simeq 1/r - \lambda$ near the nucleus) that lifts the level almost uniformly without touching the short-range relativistic well. What screening deflates is the exponential lever arm of that offset: semiclassically $\ln S \simeq -\frac{2}{3f}\left[(2|E_0|/Z^2)^{3/2} - (2|E_0^{\rm nr}|)^{3/2}\right]$, and as both levels rise toward threshold the sensitivity of the tunneling width to the persistent Δ collapses. This one-line estimate reproduces the data ($S = 0.104$ versus the measured 0.0923 at $Z = 82$, $u = 0$; 0.175 versus 0.1497 at $u = 0.20$), and freezing Δ at its unscreened value still yields $S = 0.156$ at $u = 0.20$: the level lift alone accounts for roughly 80% of the recovery, the residual outward displacement of the electron for the rest. The practical statement compounds with the known screening catalysis [3, 6]: at $Z = 82$ the absolute width grows by a factor 13.5 across the computed screening range, against 8.3 for the nonrelativistic catalysis alone — the plasma catalyzes field ionization of a heavy ion 1.6 times more strongly than nonrelativistic scaling predicts ($13.5/8.3 = 1.62$, identically the 62% recovery of S), because it simultaneously deflates the relativistic protection.

B. The $n = 2$ manifold: non-monotonic fine-structure asymmetry

At $Z = 40$ and mild scaled field $f = 0.004$ the $m = 1/2$, $n = 2$ manifold carries three states: the exactly degenerate $2s_{1/2}$–$2p_{1/2}$ pair and the $2p_{3/2}$, split by the fine structure (4.50 a.u.). An adiabatic field ramp labels the three field-dressed components by continuation — the degenerate pair splits into a downhill component A ($\Delta E = -19.65$, matching the linear Stark scale) and a central component B, while the $2p_{3/2}$-dominated C moves uphill (+13.99); labels are continuation labels, not quantum numbers. Screening continuation then gives the evolution shown in Fig. 2: the downhill component's width is catalyzed by a factor ~ 6000 across $u \leq 0.20$; the uphill component crosses $E_r = 0$ near $u \approx 0.19$ and persists as a positive-energy relativistic resonance, the Dirac analog of the through-threshold continuations of Refs. [6] (and of the field-free positive-energy states of Ref. [23]); the central, narrowest component dissolves into the mixing beyond $u = 0.05$. The relativistic observable is the width ratio of the outer pair, for which we computed the matched nonrelativistic baseline — the $n = 2$, $m = 0$ parabolic doublet at the same scaled field, with the engine of Ref. [6]: nonrelativistically the uphill/downhill width ratio starts strongly asymmetric, $\tilde{\Gamma}_{\rm up}/\tilde{\Gamma}_{\rm down} = 0.031$ at $u = 0$, and rises monotonically to 0.22 by $u = 0.15$ (beyond which the uphill state crosses threshold and its tracking degrades). The Dirac pair at $Z = 40$ resembles this baseline in neither respect. It is born with *equal* widths ($\Gamma_{\rm C}/\Gamma_{\rm A} = 1.02$ at $u = 0$): at this Z the fine-structure splitting is comparable to the Stark coupling, and the $2p_{3/2}$-dominated component borrows an order of magnitude of width through spin–orbit mixing. And its ratio is *non-monotonic*, falling to a minimum of 0.22 at $u = 0.125$ and recovering to 0.33 by $u = 0.20$ — a five-fold screening-induced asymmetry that partially heals as both states approach the continuum.

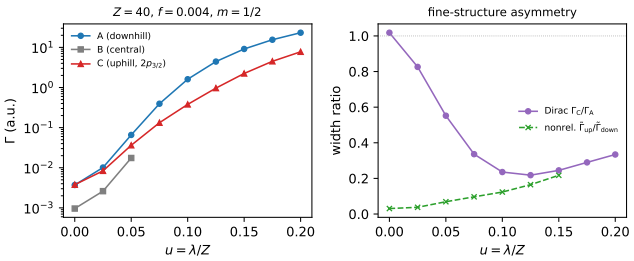


FIG. 2. The three field-dressed components of the $n = 2$, $m = 1/2$ manifold at $Z = 40$, $f = 0.004$ under screening. Left: widths (log scale) of the downhill (A), central (B) and uphill (C, $2p_{3/2}$-dominated) components. Right: the fine-structure width ratio Γ_C/Γ_A is non-monotonic in the screening parameter, while the matched nonrelativistic ratio $\tilde{\Gamma}_{up}/\tilde{\Gamma}_{down}$ (dashed; shown for $u \leq 0.15$, before the nonrelativistic uphill state crosses threshold) starts thirty-fold asymmetric and rises monotonically.

C. Honest limits

Tracking currently extends to $u = 0.15$–0.20 at $f = 0.05$ (the broadest states escape the two-angle window; the established trends are monotonic over 4–5 points per Z). At $Z = 82$ a convergence ladder (sixteen and twenty channels with $n_b = 58$ and 64) moves the $u = 0$ width by 0.3% from the production row and by only 8×10^{-5} between its last two rungs, so the row is basis-converged at the few-10^{-3} level; the converged width sits 1.6% below the value of Ref. [2], an offset that basis refinement does not close (the two calculations use different discretizations) and which we quote as the honest error bar of the extreme-Z row. QED corrections ($\sim \alpha(Z\alpha)^4$) and finite nuclear size are neglected throughout; following Ref. [2], all values are Dirac-level benchmarks. The static Debye potential is the standard model of this literature [24]; the quasistatic-microfield framing of the field follows the companion discussion [6, 25].

IV. CONCLUSIONS

The triple problem — Dirac, Debye screening, static field — is computed here, to our knowledge for the first time at the level of nonperturbative resonance positions and widths, on a validation ladder that touches every edge of the parameter cube. Its physics is a two-way coupling that neither pairwise problem contains: screening does not merely catalyze field ionization, it also erodes the relativistic suppression of the width, so that at $Z = 82$ a heavy ion in a dense plasma ionizes about 60% faster than the nonrelativistic and zero-screening treatments combined would predict; and the fine-structure asymmetry of excited Stark manifolds is a non-monotonic function of the plasma density where its nonrelativistic counterpart is monotonic. The spurious-state exclusion that made the maps possible is itself a working demonstration of the companion pollution theory [15]. Extensions within reach of the same engine include the muonic sector, finite nuclear size, and denser (Z, u, f) coverage toward line-shape applications. All scripts, gates and machine-readable maps are distributed with the companion programs and will be archived with a persistent identifier at publication.

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